

Issue Log

ISSUE-D2-01 — Transition table format implies Mealy timing, not Moore

Day found: Day 2

A literal 4-state transition table (IDLE, S1, S10, S101) is technically **Mealy-shaped** — detected depends on both state and input in the same row — even though the assignment calls for a Moore FSM.

Root cause: no state purely represents "match just completed."

Fix: added a 5th state, S1011 (DETECT), entered one cycle after the 4th matching bit; `assign detected = (state == S1011)` has no dependence on `din`, making it a genuine Moore FSM. Verified output-equivalent to a registered-Mealy alternative over 2000 random bits (0 mismatches).

Status: Resolved.

ISSUE-D3-01 — Conflated "checker" and "scoreboard" terminology

Day found: Day 3

Sources use "checker" and "scoreboard" inconsistently — a genuine field-wide ambiguity, not a personal misunderstanding.

Fix: adopted the program's own Day 10 convention (checker = expected value, scoreboard = reference model) and noted the ambiguity explicitly in the KB rather than silently picking one definition.

Status: Resolved (terminology convention fixed for this program).

ISSUE-D4-01 — Confusion over which cycle detected fires (the "cycle 8" prediction was correct all along)

Day found: Day 4

An intermediate draft wrongly "corrected" the original hand-trace (pulse one cycle after the 4th bit) to claim it fires the *same* cycle, based on a smoke log that happened to print **post-edge** values — making the pulse look simultaneous with the 4th bit. **Root cause:** a print-convention artifact, not an RTL bug.

Fix: rebuilt as a true 5-state Moore FSM (see D2-01) and switched the smoke testbench to **pre-edge** logging, which directly shows the one-cycle latency (cycle 7: state=S101; cycle 8: state=S1011, detected=1).

Status: Resolved — the original prediction was correct.

ISSUE-D5-01 — Part-select applied to a parenthesized expression (compile error)

Day found: Day 5

`(MAIN_GREEN_TIME - 1)[CNT_WIDTH-1:0]` is not legal SystemVerilog — a part-select can't apply to an arbitrary parenthesized expression, only to a signal or properly-typed constant.

Fix: removed the explicit slice and relied on normal assignment-width truncation, since the target signals were already correctly sized.

Status: Fixed. Recompiled clean with zero errors.

ISSUE-D5-02 — First `MAIN_GREEN` phase after reset is 5 cycles, not 4

Day found: Day 5 (pinned down only after fixing the logging bug in D5-03)

The reset branch loaded `count <= MAIN_GREEN_TIME` instead of `duration - 1` like every in-sequence reload, adding one extra hold cycle on the **first** phase only. This was invisible in the original log because D5-03's sampling bug dropped exactly that first cycle.

Fix: reset now loads `MAIN_GREEN_TIME - 1`, identical to the reload path; first and later `MAIN_GREEN` phases now both run exactly 4 cycles.

Status: Fixed and confirmed via re-simulation.

ISSUE-D5-03 — Traffic-light smoke testbench mis-sampled and masked the reset off-by-one

Day found: Day 5

The testbench logged one cycle late (after the first active edge), so the reset-loaded cycle never printed — the buggy 5-cycle first `MAIN_GREEN` and the correct 4-cycle later one both printed as 4 rows, hiding D5-02.

Fix: rewrote logging to sample pre-edge (registered values held *during* each cycle) with an exposed `count` column, making phase length self-evident instead of eyeballed.

Status: Fixed. General lesson: prefer self-checking or count-exposing testbenches over by-eye phase counting.

ISSUE-D6-01 — Required smoke sequence never exercises simultaneous read+write

Day found: Day 6

The specified fill-then-drain smoke sequence never asserts `wr_en` and `rd_en` in the same cycle, even though "simultaneous read+write must not change count" is a separate stated requirement — implemented logic that was untested, not verified.

Fix: extended the testbench with 3 extra simultaneous-assert cycles (full, partial, empty); all matched hand-predicted behavior exactly.

Status: Resolved — test coverage extended; RTL needed no changes.

ISSUE-D7-01 — Traffic Light minimum-duration case never actually tested

Day found: Day 7

Only the assignment's default durations (4, 2, 6, 2) had ever been simulated; the 1-cycle minimum-duration boundary case was untested.

Fix: built `traffic_light_min_duration_tb.sv` overriding `MAIN_YELLOW_TIME(1)`; confirmed the phase correctly occupies exactly one cycle with no skip or hang.

Status: Resolved.

ISSUE-D7-02 — FIFO pointer wraparound only tested once per pointer

Day found: Day 7

The existing Day 6 test wraps each pointer exactly once — a weak test of wraparound arithmetic, where an off-by-one at the wrap point could plausibly be masked on the first wrap.

Fix: built a multi-wrap testbench running 3 consecutive fill-drain cycles (24 writes, 24 reads) with automated scoreboard checking; zero mismatches across all 3 wraps of both pointers.

Status: Resolved.

ISSUE-D8-01 — Re-read pass on all 3 RTLs found 3 lingering inconsistencies

Day found: Day 8

The Day 8 re-read step surfaced three small issues sitting unfixed since earlier days:

1. `seq_detect_1011.sv` still carried a stale "cycle 7" comment from the mistaken over-correction in D4-01.
2. `traffic_light.sv` line 96 cited the wrong issue ID (ISSUE-D5-01, a syntax bug) for what is actually the D5-02 off-by-one fix.
3. `seq_detect_1011.sv` had a leftover `// Code your design here` placeholder and was missing the ``timescale` directive present in the other RTL files.

Fix: all three corrected; re-compiled with `iverilog -g2012 -Wall` — zero errors/warnings, functional results unchanged.

Status: Fixed.

ISSUE-D9-01 — Original FIFO feature list was missing a symmetric feature entirely

Day found: Day 9

The Day 3 feature list only covered `full/write-blocking`; there was no equivalent `empty/read-blocking` feature at all — not merged into anything, simply absent.

Fix: split the original feature into `full_flag_asserts_when_count_equals_depth` (CORNER_001) and `write_blocked_when_full` (NEG_001), and added the missing `empty_flag_asserts_when_count_equals_zero` (CORNER_002) and `read_blocked_when_empty` (NEG_002). Feature count: 5 → 8.

Status: Fixed.

ISSUE-D10-01 — Nearly assigned the wrong checker type by reasoning from test history instead of property type

Day found: Day 10

Initially leaned toward giving `count_unchanged_on_simultaneous_read_write` a "checker with expected value" simply because that's how it happened to be tested on Day 6 — conflating *how* a property was tested with *what kind* of property it is.

Fix: reassigned to Assertion, since the property should hold continuously on every cycle, not just at specific directed moments.

Status: Fixed (caught before finalizing the Day 10 table).

ISSUE-D12-01 — Sequence Detector and Traffic Light never received full Req-ID feature tables (traceability gap)

Day found: Day 12

The Day 9–11 verification-planning exercises (feature refinement, Req-IDs, checker/stimulus/priority, coverage) were scoped only to the FIFO — the Sequence Detector and Traffic Light never got the same treatment, meaning the D5-02 bug had no permanent tracked regression test.

Fix: built full Feature/Req-ID/Category/Checker/Stimulus/Priority/Coverage tables for both designs (6 features each), including `RST_001` as the Traffic Light's permanent regression check for D5-02.

Status: Fixed. Note: `RTL_Specifications.docx` doesn't yet reflect these tables — flagged as a known follow-up.
